

NCP-AIN AI Fabric Pro Mega Lab: Cumulus 5.4 Day-1

Bring-Up Runbook

Use this with the mixed EOS + Cumulus topology and the paste-ready Cumulus 5.4 config folder.

Boot Order

1. Import the mixed topology.
2. Boot EOS core and edge nodes.
3. Boot Cumulus leaf and border nodes.
4. Boot Ubuntu hosts only after the fabric is stable.

Cumulus Port Mapping

`swp1=Ethernet1`, `swp2=Ethernet2`, `swp3=Ethernet10`, `swp4=Ethernet11`, `swp5=Ethernet20`, `swp6=Ethernet21` on leaves or `Ethernet30` on border nodes.

Validation Focus

Check hostnames, underlay IPs, MLAG health, OSPF neighbors, EVPN neighbors, host bonds, and end-to-end traffic.

Core Commands

```
ip -br link
ip -br addr
nv show interface
bridge vlan show
clagctl

vtysh -c "show ip ospf neighbor"
vtysh -c "show bgp l2vpn evpn summary"
vtysh -c "show bgp summary"

ip addr show bond0
ip -6 addr show bond0
cat /proc/net/bonding/bond0
```

Apply Pattern

```
sudo cp <NODE>_interfaces /etc/network/interfaces
sudo cp <NODE>_frr.conf /etc/frr/frr.conf
sudo ifreload -a
sudo systemctl restart frr
```

Node Order

Apply in this order: LEAF1, LEAF2, LEAF3, LEAF4, BL1, BL2. Validate each pair after both nodes are configured.

Host Traffic Checks

```
On GPU-B:
iperf3 -s

On GPU-A:
ping -c 3 172.16.110.12
ping -c 3 -M do -s 8972 172.16.110.12
ping6 -c 3 2001:db8:110::12
iperf3 -c 172.16.110.12 -P 8
```